

ADR-001: Genesis ODE – Core vs Experimental Scope

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Project: Genesis ODE v0.4.2 and forward experimental branches

0. Purpose of This ADR

This Architecture Decision Record establishes the formal governance boundary between:

the canonical scalar Genesis ODE core, and

all experimental multiplicity, resonance, mineral, and prime-encoded extensions. The purpose is not to slow research. The purpose is to protect both sides of the work:

the scalar core remains bounded, reproducible, and defensible;

experimental branches remain free to explore without being forced to harden prematurely;

claims remain tiered, tagged, and difficult to misinterpret;

symbolic, mathematical, and physical layers remain explicitly separated.

This ADR is the controlling document for all post-v0.4.2 scope decisions unless superseded by a later ADR.

1. Context

Genesis ODE v0.4.2 introduced substrate-indexed persistence dynamics as its central architectural refinement, transitioning from a generalized coherence scalar:

```text

$C(t)$

```

to explicitly indexed substrate states:

```text

$C\_X(t)$

```

This transition stabilized the framework by separating the formal dynamical structure from substrate-specific interpretation. The v0.4.2 scalar architecture is now treated as the canonical reference implementation for substrate-indexed persistence simulation.

The scalar core has demonstrated numerical stability and interpretive usefulness across the current validation suite and example substrates, including metallurgical, semiconductor, recursive AI, organizational, semantic, and related comparative scenarios. This does not mean the scalar core is a final theory or empirically calibrated across all substrates. It means the scalar architecture is sufficiently bounded, inspectable, and reproducible to function as the current canonical engine.

Following release, several collaborative discussions and external referee reviews

identified a natural next pressure point: the scalar architecture treats each substrate as internally homogeneous. Most real substrates are not homogeneous. Metallurgical, mineral, ecological, cognitive, organizational, semantic, and AI systems often contain internal domains with distinct local stress, recovery, impedance, drag, resonance, and failure behavior.

This motivates a possible future lift:

```text

$C\_X(t) \rightarrow C\_X,i(t)$

```

where `i` denotes local microdomains within a substrate.

At the same time, exploratory discussions introduced additional future directions:

resonance injection modeling;

mineral substrate profiles;

multiplicity-aware coherence topology;

prime-encoded microdomain schemas;

local failure propagation;

structured perturbation and resonance-assisted degradation models.

These directions are promising, but they create a real architectural risk:

experimental branches may contaminate the validated scalar core, causing:

overextended claims in formal publications;

erosion of the current bounded and reproducible architecture;

symbolic inflation at the parameter, interpretation, or encoding layer;

premature hardening of exploratory ideas;

loss of adversarial tractability;

confusion between simulation behavior, physical calibration, and interpretive analogy.

This ADR establishes the governance boundary required to prevent that failure mode.

2. Decision

2.1 The Canonical Core Is Protected, Not Frozen

The Genesis ODE scalar core – as implemented in `GenesisODE\\_v0\\_4\\_2.py` / `GenesisODE\\_v0\\_4\\_2\\_rc1.py` and documented in the v0.4.2 Technical Note – is designated the current canonical engine.

The canonical core consists of:

substrate-indexed scalar persistence states `C\\_X(t)`;

substrate parameters and registry-based profiles;

impedance-aware stress propagation through `p\\_X`, `Ω\\_X`, and `D\\_k,X`;

kinematic drag accumulation;

heterogeneous cross-substrate coupling;

validation routines and failure-mode detection;

metallurgical fatigue correspondence at the scalar-substrate level;

elemental and material persistence overlays under explicit guardrails;

adversarial recursive stress testing and recursive-closure diagnostics;

expanded simulation telemetry and example runs.

The scalar core may receive:

bug fixes;

documentation refinements;

numerical-stability improvements;

validation-contract updates;

calibration examples;

backwards-compatible telemetry additions.

However, the scalar core may not be modified to accommodate experimental branch requirements without a formal superseding ADR.

Any extension that would alter the core ODE, substrate registry, impedance bridge, validation semantics, or claim boundary must instead be implemented in a separate experimental module with its own scope, test harness, evidence tier, and non-claim statement.

Rule:

> Core stability takes precedence over experimental convenience.

2.2 Experimental Branches Are Explicitly Quarantined

The following work streams are designated EXPERIMENTAL and must be developed in clearly separated modules, directories, and documentation:

Branch	Designation	Repo Path / Artifact Path
Multiplicity-aware persistence	EXPERIMENTAL	`GenesisODE\Multiplicity\Experimental\`
Resonant injection modeling	EXPERIMENTAL	`GenesisODE\Resonance\Experimental\`
Mineral substrate profiles	EXPERIMENTAL	`GenesisODE\Mineral\Experimental\`
Prime-encoded microdomain schemas	EXPERIMENTAL	`PRIME\_SCHEMA\_*\`
Resonant mineral-coherence experiments	EXPERIMENTAL	`Resonant\Mineral\Coherence\Experimental\`

Experimental branches:

may not be cited as validated Genesis core behavior;

may not inherit evidentiary authority from the scalar core by proximity, naming, or shared notation;

must carry explicit `EXPERIMENTAL` headers in documents and source files;

must maintain their own test harnesses, failure-mode notes, and non-claim sections;

must declare decomposition basis, evidence tier, and intended use before external distribution;

must avoid public language implying calibration, physical confirmation, or universal mechanism unless such support is explicitly documented.

These branches are also explicitly protected as the project's innovation budget.

They may explore high-uncertainty ideas, but they must remain clearly marked as exploratory and may not be pressured into publication-ready claims before calibration, validation, and governance structures are mature.

Rule:

> Experimental space is protected, but experimental claims are not promoted by proximity to the core.

2.3 Branch Promotion Criteria

An experimental branch may only be proposed for promotion into the canonical core if all of the following are satisfied:

Declared Scope: The branch has a clear README or design note stating its purpose, boundaries, and non-claims.

Evidence Tier: All substrate claims are tiered under the evidence taxonomy in Section 2.5.

Validation Contract: The branch has executable or explicitly reproducible tests.
 Layer Tags: Core claims are tagged `[E]`, `[S]`, or `[I]` under Section 2.6.
 No Unresolved Ambiguity: Any unresolved core claim ambiguity has been downgraded to `[I]` or held pending redesign.
 Calibration or Benchmark: At least one relevant calibration path, benchmark, or null-model comparison exists.
 Referee Review: Phase Mirror or equivalent external review has not identified unresolved layer bleed.
 Superseding ADR: Promotion requires a new ADR or explicit amendment to this ADR.
 Until these conditions are met, experimental branches remain exploratory regardless of how promising their outputs appear.

2.4 Prime-Encoding Discipline

This clause is the constitutional firewall for all multiplicity-related work.

The governing rule is:

> In the multiplicity-experimental branch, decomposition is first into physically legible microdomains. Prime structure enters only as a semantically loaded encoding of those microdomains under canonical integer factorization. `\**Dynamics` live on `C_X,i`, not on prime labels. `\**`

This means:

Decomposition is physical and substrate-readable. Examples include grain families, fracture zones, inclusions, hydration regions, weld regions, heat-affected areas, microcrack clusters, or equivalent substrate-legible partitions.

Encoding is mathematical. Each microdomain `i` receives an integer code `n_i` whose prime factorization acts as a canonical, typed, reversible descriptor:

````text`

$$n_i = \prod p_k^{e_{i,k}}$$

`````

Dynamics evolve on local state variables, including:

````text`

`C_X,i`

`p_X,i`

`Ω_X,i`

`D_k,X,i`

`f_n,i`

`Q_i`

local thresholds

local recovery terms

`````

Prime factors inform grouping, telemetry, schema decoding, feature comparison, and coupling heuristics only.

Prime labels are never dynamical coordinates.

The integer is structured metadata, not a physical mode.

Example: Metallurgical Grain Family Encoding

A grain family might be encoded as:

````text`

$$n_i = 2^3 \times 5^2 \times 13^1$$

`````

where the primes map to documented semantic bands and the exponents are bounded ordinals or count-like intensities.

For example:

| Prime Band | Example Meaning | Example Exponent | Interpretation |
|------------|------------------|------------------|--|
| `2^a` | composition band | | alloy phase / mineral family intensity |
| `5^b` | geometry band | | grain size or aspect-ratio class |
| `13^c` | defect band | | microcrack-density class |

This example is illustrative only. The actual assignments must be defined in `PRIME\Schema\Metallurgical.md` before any prime-encoded multiplicity work is distributed.

Metallurgical Prime Band Families

The initial metallurgical schema shall reserve semantic bands for:

Semantic Family Encoding Role

Composition band Material constitution: alloy phase, mineral family, impurity class

Geometry band Morphology: grain size class, aspect ratio, orientation band

History band Processing lineage: heat treatment, weld status, cold-work, hydration history

Defect band Local flaw structure: inclusion type, porosity class, microcrack density

Failure-signature band Empirical risk markers: fatigue tendency, brittle fracture propensity, delamination risk

Each prime in a band must have:

- a fixed documented meaning;

- a bounded exponent range;

- an allowed interpretation;

- a disallowed interpretation;

- at least one example;

- a note indicating whether it is empirical, structural, or interpretive.

`PRIME\Schema\Metallurgical.md` is the control surface that prevents encoding drift.

2.5 Evidence Tier Commitment

All substrate classes must carry an explicit evidence tier before use in comparative analysis or external publication.

| Tier | Designation | Meaning |
|------|-------------|---------|
|------|-------------|---------|

| | | |
|---|--------------------------|--|
| A | Empirically calibratable | Aims to track or qualitatively match known curves, data, or behaviors from domain literature |
|---|--------------------------|--|

| | | |
|---|-----------------------|---|
| B | Stylized but anchored | Structurally informed by domain literature; not yet calibrated against a specific dataset |
|---|-----------------------|---|

| | | |
|---|---------------------------------------|---|
| C | Explicitly metaphorical / exploratory | Heuristic or pedagogical use only; no empirical grounding claimed |
|---|---------------------------------------|---|

Substrates may not be rhetorically conflated across tiers. A Tier A metallurgical scenario and a Tier C elemental overlay are not interchangeable analytical claims. The artifact `SUBSTRATES\EVIDENCE\TIERS.md` is the highest-priority companion document to this ADR. It must specify for each substrate:

- substrate name and symbol;

- intended use;

- observables or proxy variables;

- evidence tier;

- admissible decomposition basis;

- disallowed interpretations;

current calibration status;
whether the substrate is canonical, local-extension, or experimental.

Rule:

> A substrate without an evidence tier is not publication-ready.

2.6 Layer Separation Rule

All claims produced within the Genesis framework must be tagged by epistemic layer before publication or external distribution.

| Tag | Layer | Description |
|--------|--------------|---|
| `\[E]` | Empirical | Calibrated against data, directly measured, or explicitly benchmarked |
| `\[S]` | Structural | Derived from formal architecture or simulation behavior under stated assumptions |
| `\[I]` | Interpretive | Analogical, metaphorical, pedagogical, or heuristic; not a formal mechanism claim |

Any claim that remains unresolved after structured layer review must be either: downgraded to `[I]`, or

held pending structural redesign.

There is no valid publication state in which a core claim remains ambiguously tagged.

Rule:

> Ambiguity is allowed in exploration. It is not allowed to harden silently into a core claim.

2.7 Protocol-as-Microscope Rule

Firewalker, Phase Mirror, internal red-team review, validation scripts, and related governance tools may be cited only as process methods, never as evidence that a scientific or technical claim is true.

They may support statements such as:

```text

This claim was reviewed under Firewalker / Phase Mirror process.

```

They may not support statements such as:

```text

This claim is true because Firewalker / Phase Mirror approved it.

```

Rule:

> Governance tools expose failure surfaces; they do not certify truth.

3. Consequences

3.1 What This Protects

This ADR protects:

the scalar Genesis core as a bounded, tractable, reproducible reference architecture;

formal publications grounded in v0.4.2 scalar behavior;

experimental freedom without premature claim hardening;

substrate-indexing discipline;

future multiplicity work from numerological or symbolic overreach;

public trust in the framework's evidence boundaries.

3.2 What This Requires

All new Genesis-related modules must declare:

Core or Experimental status;
evidence tier;
layer tag discipline;
claim boundary;
test harness or validation status;
non-claims.

No experimental result may be cited as “Genesis behavior” in a formal publication without explicit branch and evidence-tier annotation.

Required artifacts:

| Artifact | Priority | Due Date / Horizon |
|---|----------|--------------------|
| `SUBSTRATES\_EVIDENCE\_TIERS.md` | | Highest 7 days |
| `VALIDATION\_CONTRACT.md` | High | 14 days |
| `METALLURGICAL\_CALIBRATION.md` | High | 30 days |
| `PRIME\_SCHEMA\_Metallurgical.md` | High | 30 days |
| `RESONANT\_INJECTION\_APPENDIX.md` | Medium | 30-60 days |
| `GenesisODE\_Multiplicity\_Experimental/microdomain\_stub.py` | Medium | 30-60 days |

4. Rollback Triggers

If any of the following conditions are observed, this ADR must be reviewed and the affected experimental branch must be suspended pending re-evaluation:

An experimental branch claim appears in a formal publication without explicit tier annotation.

Prime labels are used as dynamical drivers rather than metadata in any simulation output.

An experimental module is cited as validation evidence for a scalar core claim.

Repeated layer-classification disputes in peer review suggest `[E] / [S] / [I]` tagging is not functioning as a governance gate.

Phase Mirror review identifies systematic layer bleed.

Public-facing materials imply resonance, mineral, or prime-encoded work is validated core Genesis behavior.

A governance protocol is cited as evidence rather than process.

Experimental branch outputs create pressure to alter the scalar core without a superseding ADR.

Rollback authority rests with the Lead Architect, with formal advisory input from the Phase Mirror referee layer.

Any rollback event must be logged with:

trigger identified;
branch or artifact affected;
suspension action taken;
corrective action required;
reviewer notes;
date of restoration or supersession.

5. Artifact Dependency Map

```text

|                                           |                              |
|-------------------------------------------|------------------------------|
| ADR-001\_Core\_vs\_Experimental\_Scope.md |                              |
| └─ SUBSTRATES\_EVIDENCE\_TIERS.md         | \[7 days – highest priority] |
| └─ VALIDATION\_CONTRACT.md                | \[14 days]                   |
| └─ METALLURGICAL\_CALIBRATION.md          | \[30 days]                   |

```

├── PRIME_SCHEMA_Metallurgical.md \[30 days]
├── RESONANT_INJECTION_APPENDIX.md \[30-60 days]
├── GenesisODE_Multiplicity_Experimental/
│ ├── README.md \[30-60 days]
│ ├── microdomain_stub.py \[30-60 days]
│ └── tests/
...

```

All artifacts downstream of this ADR inherit its claim-boundary rules. An artifact that contradicts ADR-001 does not supersede it; it requires a formal ADR revision with documented rationale and referee review.

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## 6. Required Experimental Branch Header

Every experimental document and source file must begin with a header substantially equivalent to:

```

```text
STATUS: EXPERIMENTAL
BRANCH: <branch name>
CORE STATUS: Not part of canonical Genesis scalar core
EVIDENCE TIER: <A/B/C or mixed>
CLAIM BOUNDARY: Exploratory; not validated Genesis core behavior
NON-CLAIMS: <explicit disallowed interpretations>
```

```

For code files, this may appear as a module docstring. For Markdown, it must appear before the first substantive section.

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## 7. Disallowed Interpretations

The following interpretations are explicitly disallowed unless a future superseding ADR and evidence package authorize them:

Mineral resonance experiments do not demonstrate ancient soft-stone technology.

Resonant injection terms do not imply transmutation or exotic physics.

57.8 GHz or any other carrier frequency is not a validated universal eigenfrequency within Genesis ODE.

Prime-encoded microdomain schemas do not imply primes are physical microdomains.

Multiplicity-aware behavior does not invalidate the scalar core.

Experimental branch stability does not validate core Genesis claims.

Scalar core validation does not validate experimental branch claims.

Elemental or symbolic overlays do not replace chemistry, metallurgy, mineralogy, neuroscience, psychology, or established physics.

Recursive AI persistence metrics do not imply AI consciousness.

Semantic coherence metrics do not imply objective truth.

Governance approval does not imply empirical confirmation.

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## 8. What This ADR Does Not Claim

This ADR does not claim:

that multiplicity or resonance directions are closed;

that the scalar core is the final form of Genesis;

that experimental branches are invalid;

that prime-encoding is required for all future work;

that all substrates can be calibrated equally;

that Genesis ODE is a universal ontology;



that governance tools determine truth.

This ADR permits exploratory discussion and conceptual development when clearly marked as exploratory.

The architecture advances by becoming more executable, more falsifiable, more typed, and more difficult to misinterpret – not by expanding its conceptual reach.

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## 9. Review and Amendment

This ADR is open for Phase Mirror review and amendment.

Proposed amendments must include:

reason for change;

affected section;

claim-boundary impact;

evidence-tier impact;

reviewer notes;

whether the amendment weakens or strengthens Sections 2.1–2.4.

No amendment may reduce the claim-boundary protections in Sections 2.1–2.4 without a superseding ADR.

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## 10. Signatures

| Role                 | Name                             | Date       | Status         |
|----------------------|----------------------------------|------------|----------------|
| Lead Architect       | Allan Christopher Beckingham, CD | 2026-05-15 | APPROVED       |
| Phase Mirror Referee | Dr. Ryan van Gelder              | –          | PENDING REVIEW |
| CDL Review Layer     | Athena / Zen / Limen             | 2026-05-15 | ADVISORY INPUT |

INCORPORATED

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## 11. Change Log

Version Change

1.0 Initial Limen draft establishing core-vs-experimental boundary

1.1 Athena refinements: innovation budget, prime-schema example, evidence-tier priority, Phase Mirror rollback trigger

1.2 Consolidated version: added core-not-frozen clause, branch promotion criteria, protocol-as-microscope rule, required experimental header, disallowed interpretations, review/amendment procedure, and expanded rollback triggers

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Genesis ODE Project – Governance Layer

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